

VARUN PATIL

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EDUCATION

International Institute of Information Technology, Hyderabad

Master of Technology in Product Design and Management

Hyderabad, TG

Aug. 2025 – Present

JSPM's Rajarshi Shahu College of Engineering

Bachelor of Technology in Computer Science Engineering

Pune, MH

Aug. 2020 – June 2024

EXPERIENCE

Software Engineering Intern

Jan. 2024 – July 2024

Data Axle

Pune, MH

- Developed and maintained backend services and web applications using **C#, ASP.NET Core, SQL Server, and React**.
- Built REST APIs and data-processing workflows, improving application performance through query optimization and efficient database design to support customer acquisition, business intelligence, and data management platforms.
- Improved software delivery through **CI/CD enhancements**, automated testing, and **Azure DevOps-based** deployment pipelines.

PROJECTS

Atelier | WebSockets, CRDTs, Distributed Systems, PostgreSQL, Prometheus

GitHub

- Built a real-time collaborative code editor supporting **20+ concurrent users** with **sub-50 ms synchronization latency** using CRDTs (Yjs) based synchronization, distributed WebSocket gateways, and offline conflict resolution.
- Designed an **offline-first collaboration** model enabling local edits to be merged automatically after reconnection.
- Implemented **live presence tracking** and cursor synchronization to provide a shared editing experience.
- Implemented remote code execution by spawning isolated pseudo-terminals (node-pty) on the server, piping stdin/stdout over WebSockets to an xterm.js terminal, supporting JavaScript, Python, C++, and Java
- Engineered a time-travel replay player that rebuilds a room's exact state at any moment in history by folding a recorded CRDT event log – reconstructing the document (in a read-only Monaco editor), who was present, changes to the editor.
- Designed an AI agent system where agents join as live participants – retrieval-grounded, human-approved edits with reasoning visible in real time and in replay.

StreamRank | PyTorch, Apache Kafka, Two-Tower Networks, FAISS, Kubernetes

GitHub

- Designed and built a TikTok-style recommendation engine that takes a user's watch history and returns a personalized video feed in real time, using a four-stage funnel — retrieve candidates, rank them, re-rank for diversity, and serve the top 10 — trained on 12.5M real user interactions and served through a REST API at 3ms per request.
- Developed the core ranking model as a sequential transformer that reads a user's recent watch behavior in order (not as a bag of features), predicting both whether they'll engage and how long they'll watch — outperforming a gradient-boosted-tree baseline by a wide margin on watch-time prediction.
- Built the real-time layer so the feed adapts within a session: user interactions stream through Kafka into a consumer that continuously updates live features in Redis, so a video watched now reshapes the very next set of recommendations.
- Added a contextual bandit that learns to balance engagement against feed diversity and freshness, and validated its impact with off-policy evaluation — a causal-inference technique that estimates how a new policy would have performed from logged data, without risking live users.

TECHNICAL SKILLS

Languages Java Python C/C++ C# JavaScript

Frameworks React ASP.NET Core Node.js Flask JUnit RestAPI FastAPI

Machine Learning PyTorch Deep Learning LightGBM DeepFM Two-Tower Networks

Distributed Systems Apache Kafka Redis FAISS Docker FastAPI WebSockets CRDTs

CS Fundamentals OS DBMS Computer Networks Object Oriented Design Data Structures & Algorithms

ACHIEVEMENTS

- Secured **AIR 202** in **PGEE CSE 2025**, qualifying for admission to IIIT Hyderabad.
- Achieved **AIR 2849** in **GATE CS 2025** among 170,000+ candidates, corresponding to the **98.14 percentile**.